

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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August 26, 2025

Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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§Thanks to Nick Bloom, Joshua Gans, David Autor, Daniel Rock, Fei-Fei Li, Frank Li, Christina Langer, Sarah Bana, Cody Cook, Chris Forman, Andrew Wang, Brad Ross, Omeed Maghzian, Basil Halperin, Jiaxin Pei, Phil Trammell, Eric Bergman and participants at the Stanford Digital Economy Lab workshop for helpful feedback. We are grateful to ADP for access to the data and the Stanford Digital Economy Lab for financial support. All errors are our own.

¶Latest version: <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>

granularity, providing a near real-time view of labor market adjustments. By linking this data to established measures of occupational AI exposure and other variables, we can quantify the realized employment changes since the widespread adoption of generative AI.

This paper systematically presents six key facts that emerge from the data, offering an assessment of how the AI revolution is reshaping the American workforce.

Our first key finding is that we uncover substantial declines in employment for early-career workers (ages 22-25) in occupations most exposed to AI, such as software developers and customer service representatives. In contrast, employment trends for more experienced workers in the same occupations, and workers of all ages in less-exposed occupations such as nursing aides, have remained stable or continued to grow.

Our second key fact is that overall employment continues to grow robustly, but employment growth for young workers in particular has been stagnant since late 2022. In jobs less exposed to AI young workers have experienced comparable employment growth to older workers. In contrast, workers aged 22 to 25 have experienced a 6% decline in employment from late 2022 to July 2025 in the most AI-exposed occupations, compared to a 6-9% increase for older workers. These results suggest that declining employment AI-exposed jobs is driving tepid overall employment growth for 22- to 25- year-olds as employment for older workers continues to grow.

Our third key fact is that not all uses of AI are associated with declines in employment. In particular, entry-level employment has declined in applications of AI that *automate* work, but not those that most *augment* it. We distinguish between automation and augmentation empirically using estimates of the extent to which observed queries to Claude, the LLM, substitute or complement for the tasks in that occupation. While we find employment declines for young workers in occupations where AI primarily automates work, we find employment growth in occupations in which AI use is most augmentative. These findings are consistent with automative uses of AI substituting for labor while augmentative uses do not.

Fourth, we find that employment declines for young, AI-exposed workers remain after conditioning on firm-time effects. One class of explanations for our patterns is that they may be driven by industry- or firm-level shocks such as interest rate changes that correlate with sorting patterns by age and measured AI exposure. We test for a class of such confounders by controlling for firm-time effects in an event study regression, absorbing aggregate firm shocks that impact all workers at a

Both the [Eloundou et al. \(2024\)](#) measures and the [Handa et al. \(2025\)](#) measures estimate AI exposure by 2018 SOC code. We use a 2010 SOC code to 2018 SOC code crosswalk from the BLS to merge the exposure measures to the payroll data. Table [A1](#) shows example occupations for each AI exposure measure.

3.3 Other Data

To compare employment changes for teleworkable versus non-teleworkable occupations, we use data from [Dingel and Neiman \(2020\)](#). We use the Personal Consumption Expenditure index from the BLS to compute real earnings, indexed to October 2017. We use monthly Current Population Survey (CPS) data as a comparison for our main findings.

4 Results

4.1 Fact 1: Employment for young workers has declined in AI-exposed occupations

Consider software engineers and customer service agents, two occupations that are frequently considered to be highly exposed to generative AI tools. Media attention has raised the specter of widespread employment disruption for young software engineers in particular ([Thompson, 2025](#); [Raman, 2025](#); [Allen, 2025](#); [Horowitch, 2025](#)).

Figure [1](#) shows employment changes by age group for these occupations, normalized to 1 in October 2022. Both occupations present a similar pattern: employment for the youngest workers declines considerably after 2022, while employment for other age groups continues to grow. By July, 2025, employment for software developers aged 22-25 declined by nearly 20% compared to its peak in late 2022. Figure [A1](#) shows that a similar pattern holds for computer occupations and service clerks more generally.

Figure [2](#) shows four other professions as case studies, spanning varying levels of AI exposure according to the measures developed in [Eloundou et al. \(2024\)](#). Marketing and sales managers, in

refinement process; Learning, meaning knowledge acquisition and understanding; Validation, meaning work verification and improvement; or “None,” with the model instructed to choose the None option “liberally.” Conversations classified as Directive or Feedback Loop are considered as Automative, while ones classified as Task Iteration, Learning, or Validation are considered Augmentative. See [Handa et al. \(2025\)](#) for more details. Table [A2](#) reproduces Table 1 from [Handa et al. \(2025\)](#) and shows more details about the automation and augmentation measures.

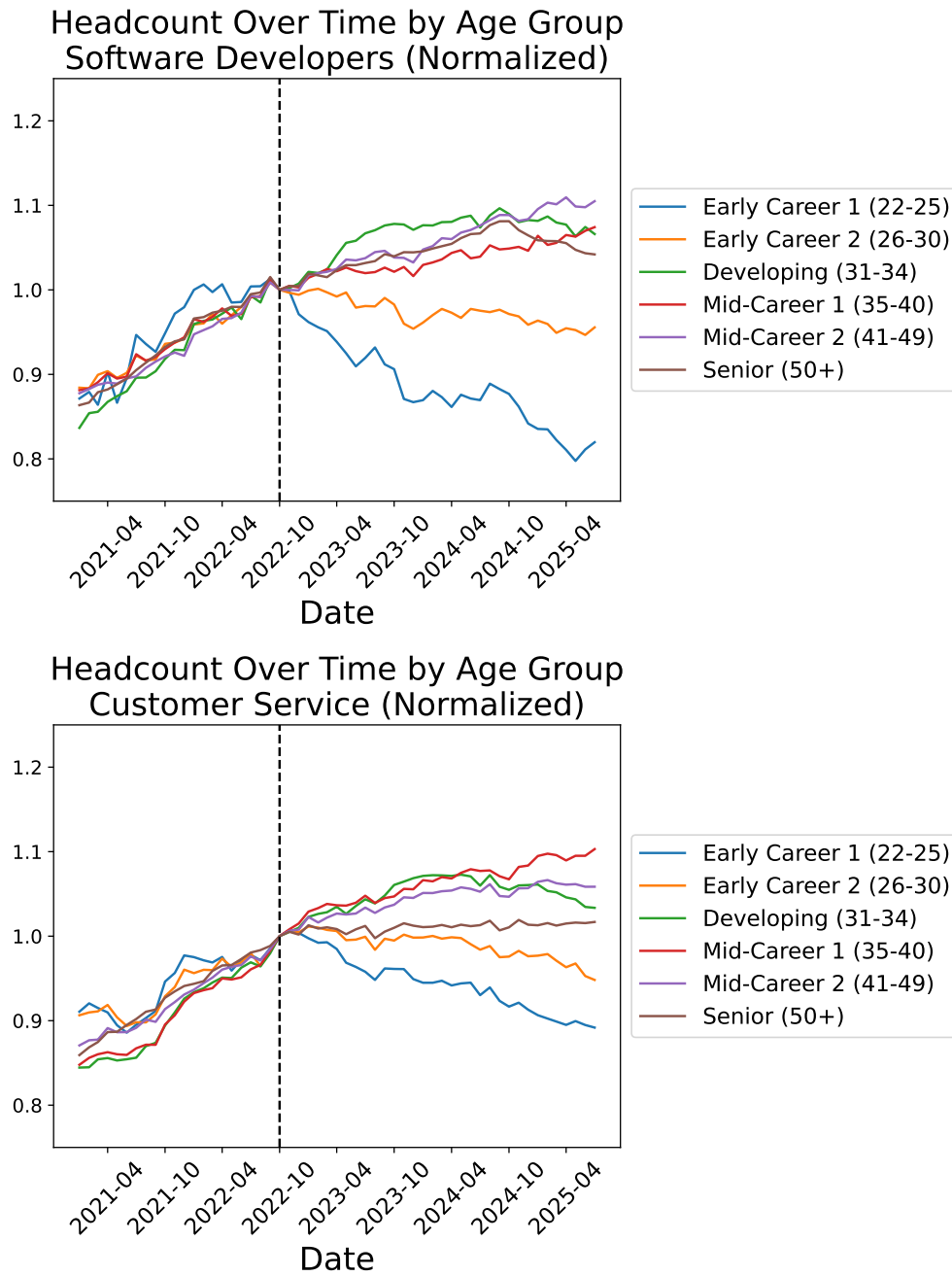


Figure 1: Employment changes for software developers and customer service agents by age, normalized to 1 in October 2022.